

Ian Cosgrove

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Education

University of California, San Diego

San Diego, CA

M.S., Mechanical Engineering

Expected Oct. 2026

- Thesis: Turbulent Dissipation in the Bottom Boundary Layer of La Jolla Canyon
- Cumulative GPA: 4.0/4.0

B.S., Mechanical Engineering, *cum laude*

Sept. 2021 - June 2025

- Focus: Environmental & Computational Fluid Dynamics, Renewable Energy
- Cumulative GPA: 3.938/4.0

Technical Skills

Programming: MATLAB (signal detection algorithms, time-series & spectral analysis, UI development, numerical methods, quality control, data visualization)

Simulation and CAD: Ansys Fluent, Ansys Mechanical, Fusion 360, SolidWorks

Certifications: Engineer-in-Training (Mechanical), Ansys Fluent: Fluid Dynamics & Compressible Flow, Ansys Mechanical: Heat Transfer & Stress Analysis

Instrumentation: Nortek & RDI Acoustic Doppler Current Profiler, Seabird Thermistor & CTD, HII Remus 100 autonomous underwater vehicle

Experience

Graduate Student Researcher: Environmental Fluid Dynamics

Dec. 2024 - Present

UC San Diego Mechanical and Aerospace Engineering Department

- Processed and analyzed oceanographic datasets (ADCP, thermistor, CTD) to calculate turbulent dissipation in the bottom boundary layer of the La Jolla submarine canyon.
- Developed MATLAB programs that apply multiple dissipation methods: inertial subrange fitting, Thorpe scale analysis and structure functions for independent validation.
- Interpreted spectral and spectrogram analyses, space/time-averaged statistical results, and created advanced data visualizations to characterize tidally-driven turbulent bores.
- Operated a Remus 100 autonomous underwater vehicle in a submarine canyon, analyzed mission data for target variables (dye, temperature) and anomalies.

Student Researcher: Underwater Acoustics

June 2023 - June 2026

UC San Diego Scripps Institution of Oceanography

- Developed a MATLAB UI to rapidly review user-flagged acoustic signals and gather acoustic parameters (SNR, bandwidth). Utilized to catalog low-frequency signals at a new study site.
- Expanded an automatic signal detection algorithm with tools to compute received level and background noise level, identified parameters for operation in a high ambient-noise site.
- Designed, executed, and analyzed laboratory tests of a physical hydrophone mooring setup to reduce self-imposed noise and improve data quality in acoustically complex sites.

Project

Novel Carbon Dioxide Capture Device: Senior Capstone Project

Feb. 2025 - June 2025

- Calculated geometric and performance criteria for a passive-diffusion air-to-NaOH solvent carbon capture device to meet CO₂ removal goals.
- Performed CFD simulations using Ansys Fluent to model internal solvent and external air flow.
- Designed, executed, and analyzed prototype performance tests.